

Morgan Youngblood Curriculum Vitae

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USCG Master 25GT w/ Towing Endorsement

Application in Review: January, 2026

Education

Bachelor of Science in Marine Science, University of Hawai'i at Hilo
Pikes Peak Community College at Colorado Springs

August 2021-May 2025
August 2019-May 2021

Honors, Awards, & Accomplishments

National Science Foundation GRFP Honorable Mention (2025)
STEM Research Honors Certificate Program (2025)
Marine Option Program Certificate Specialty in Water Quality Research (2025)
President, Marine Technology Society, University of Hawai'i Hilo Chapter (2025)
College of Natural and Health Sciences Excellence Award in Marine Science (2024)
University of Hawai'i Marine Option Program Certificate (2024)
Dean's List (2023-2025)
National Oceanic and Atmospheric Administration Ernest F. Hollings Scholarship (2023)

Relevant Coursework

- Small Boat Operations & Research
- Oceanography and Lab
- Marine Biology-Diversity and Lab
- Statistical Apps in Marine Science
- Applied Informatics
- Ecology and Evolution
- Writing for Science and Technology
- Teaching Marine Science
- Calculus 2
- Global Change
- Geology
- General Chemistry 2 and Lab
- Organic Chemistry 2 and Lab
- Physics 2 and Lab
- Pelagic Methods and Analysis and Lab
- Geological Oceanography
- Chemical Oceanography
- Physical Oceanography
- Public Speaking

Research Experience

Modeling Fish Community Structure in Upwelling Driven Systems

August 2024-Present

University of Hawai'i Hilo Senior Thesis Capstone Project

- Developed interactive Google Earth Engine applications to visualize multi-decadal datasets
- Architected automated R-scripts to synthesize multi-decadal biogeophysical datasets
- Implemented rigorous QA/QC protocols across cross-sector scientific databases.
- Forecasted fish community dynamics in response to environmental variables
- Implemented geospatial techniques to visualize community distribution patterns
- Generated insights to inform ecosystem-based management strategies

Apex and Mesopredator Habitat Fidelity and Movement Analysis

May 2024-May 2025

NOAA Hollings Internship Channel Islands National Marine Sanctuary

- Executed QA/QC and synthesis of decade-long Innovasea acoustic telemetry datasets
- Engineered R-based data pipelines to optimize inter-organizational big data management
- Managed cross-sector data-sharing protocols with West Coast species monitoring networks
- Authored technical manuscript for the National Marine Sanctuary network
- Assisted with on-board equipment retrieval and deployment

Kona Coast Water Quality Assessment Project

January 2023-May 2024

Marine Option Program (MOP) Student Lab and Field Assistant

- Conducted precision field sampling and laboratory analysis of coastal water quality parameters
- Cultured *Enterococcus faecalis*
- Performed geospatial analysis using ArcGIS
- Performed regulatory compliance analysis against State of Hawai'i public health standards
- Calibrated and operated precision instruments: YSI, Turbidimeter, pH Meter

- Prepared samples for Chlorophyll a, $\delta^{15}\text{N}$ -, and $\delta^{18}\text{O}$ - NO_3 - analysis

Presentations

Fish Assemblages in Wind-Driven Upwelling Zones in California: A Long-Term Perspective

AFS Inaugural meeting for the Pacific Islands chapter

April 2025

Honolulu, Hawai'i

Fidelity and Flux: Comparative Analysis of Marine Predator Residency

AAUS Diving for Science Symposium

March 2025

Seattle, Washington

Western Society of Naturalists 105th Annual Meeting

November 2024

Portland, Oregon

NOAA Office of Education Science and Education Symposium

August 2024

Silver Spring, Maryland

Investigating Environmental Health and Sustainable Wastewater

Infrastructure In Kailua-Kona, Hawai'i, USA

University of Hawai'i Marine Option Program Symposium

April 2024

Hilo & Kahului, Hawai'i

Contributions

Presentation:

Title: Satellite and Acoustic Telemetry Reveals Deep Dives and Long Range Movements of the Bat Ray (*Myliobatis californica*)

July 2024

Role: Contributing Author

Speaker: Dr. Ryan Freedman

Publication:

Title: Predicting Sea Level Rise Impacts on Coastal Wastewater Infrastructure and Water Quality for Adaptive Planning and Increased Coastal Habitat Resilience

January 2023-December 2024

Role: Technical Field and Lab Assistant

Mentors: Dr. Tracy Wiegner, Dr. Steven Colbert, Ihilani Kamau

Work Experience

Hilo Ocean Adventures

August 2022-Present

Captain/ Marine Operations Lead

- Execute commercial maritime operations for high-cadence marine excursions
- Captain and lead crew for contracts serving up to 6 passengers
- Implement safety protocols, pre-departure checks, and hazard assessments
- Manage operations, logistics, and briefings for scientific and recreational charters
- Mentor and direct junior crew in marine safety and field protocols

UH Hilo Boat Program

November 2022-August 2025

Instructional/ Research Assistant

- Assisted captain during instructional or research cruises
- Deployed sampling gear: YSI, Plankton Net, Core Sampler, Niskin Bottle, Light Meter, Castaway CTD
- Deployed remote sensing equipment: PACIOOS Water Quality Monitoring Buoy
- Performed boat maintenance and preparation
- Handled lines

UH Hilo Marine Science Department

August 2023-May 2025

Pelagic Methods and Analysis Field Lab Assistant

- Collaborated with professors on a weekly basis
- Prepared/calibrated sampling equipment
- Printed and reviewed coursework materials
- Transported students and gear to/from the field site
- Aided students in boat, lab, and field work
- Maintained order and cleanliness in department facilities

Pikes Peak Bike Tours, Colorado Springs, CO

March 2018-August 2022

Business Development Associate & Lead Guide

- Expanded local business relations and partnerships
- Explored untapped markets through product development
- Prioritized quality of service and ethical use of natural resources
- Managed guide hiring and training protocols
- Oversaw backcountry tours for corporate groups of up to 50

Sea Quest, Kona, Hawai'i

December 2016-November 2017

In-Water Boat Crew

- Accrued extensive boat time
- Operated in a variety of conditions, including night, low visibility, and high surf
- Conducted manta ray snorkel tours
- Crewed snorkel excursions in historic and culturally significant locations

Kona Boys, Hawai'i

October 2016-March 2017

Ocean Guide

- Steered and directed *W'a'a* Hawaiian outrigger canoe tours
- Led kayak snorkel tours in historic and culturally significant locations
- Instructed stand-up paddle boarding lessons

Technical Skills

Data Analysis & Software

- Acoustic Telemetry Data Synthesis (Innovasea)
- Acoustic Telemetry: Innovasea VR2X, OTN network protocols
- Google Earth Engine: JavaScript API, ui.Panel, live app deployment
- Statistical Analysis: R (sf, terra, ncdf4, ggplot2, mgcv, FSSgam, MuMIn, tidyverse)
- Geospatial Analysis: ArcGIS Pro, ArcGIS Online, Python (geopandas, pandas)
- Remote Sensing: MODIS, NOAA CDR Pathfinder, ETOPO1, PacIOOS/PIRAT
- Version Control: Git, GitHub (sci-youngblood)
- Microsoft Office Suite: Word, Excel, PowerPoint

Laboratory Techniques

- Spectrophotometry: Silica, Chlorophyll a analysis
- Total alkalinity analysis through titration
- Sample preparation for Chlorophyll a, $\delta^{15}\text{N}$ -, and $\delta^{18}\text{O}$ -NO₃- analysis

Marine Operations & Field Work

- ROV & USV field observation (Blue Robotics)
- Launch and recovery systems
- Powerboat operations and maintenance
- Deployment and retrieval of buoyed remote sensing technology
- Water quality monitoring using precision equipment
- Calibration of sampling devices: YSI, Refractometer, Turbidimeter, pH Meter

Certifications

PADI Advanced Open Water

Completed: October 2023

PADI #: 23040D9295

Lifeguarding with CPR/AED for Professional Rescuers and First Aid

Completed: August 2025

Conducted By: Red Cross

Certificate ID: 012DDQ6

Wilderness First Responder

Completed: March 2019

Conducted By: Wilderness Medical Associates International

Course ID: 29286